1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer, high intratumoral TIL density does not predict pembrolizumab response when tumors harbor oncogenic KRAS mutations because KRAS-driven signaling programs a locally immunosuppressive microenvironment that impairs CD8+ T cell effector function and diminishes PD-1 blockade efficacy; TP53 mutations modulate this effect by reshaping the neoantigen landscape and antigen presentation context. Therefore, among TIL-high MIBC tumors, KRAS-mutant status will be associated with a lower objective response rate to pembrolizumab compared with KRAS-wildtype tumors, with TP53 mutation status providing context-dependent modulation of response via changes in neoantigen burden. This yields testable predictions: (1) the ORR is lower in TIL-high/KRAS-mutant vs. TIL-high/KRAS-wildtype tumors; (2) the magnitude of this effect is modulated by TP53 status (mutant vs. wild-type) due to differential neoantigen load and recognition potential.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	KRAS, TP53
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS because oncogenic KRAS signaling is mechanistically poised to reprogram the tumor microenvironment toward immunosuppression and impair T cell function; I included TP53 as a second anchor because TP53 mutation status is tightly associated with urothelial carcinoma and can reshape the tumor mutational/antigen landscape.

TP53: TP53↔urothelial carcinoma association suggest DNA-damage–driven changes in neoantigen burden and genetic context that could modulate immune recognition and checkpoint efficacy.

3. Clinically Testable Predictions

Prediction 1: and tissue for TIL scoring; primary analysis uses objective response rate (ORR) as the endpoint, comparing TIL-high/KRAS-mutant vs TIL-high/KRAS-wildtype, with TP53 status as a stratifying/modulating factor. Include covariates such as PD-L1 expression and tumor mutational burden to control for known confounders.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions